

Welcome to 0xHACK'26

OxHack is Back!



350 hackers. 2 days. 1 event.

- OxHack 2014, ...
- OxfordHack 2018, 2019, 2020, 2022
- Various other large scale events

Team

We're wearing OxHack fleeces!

- Raphael
- Kaal
- Rayn
- Declan
- Anshul
- Amos
- And others...

OxHack26

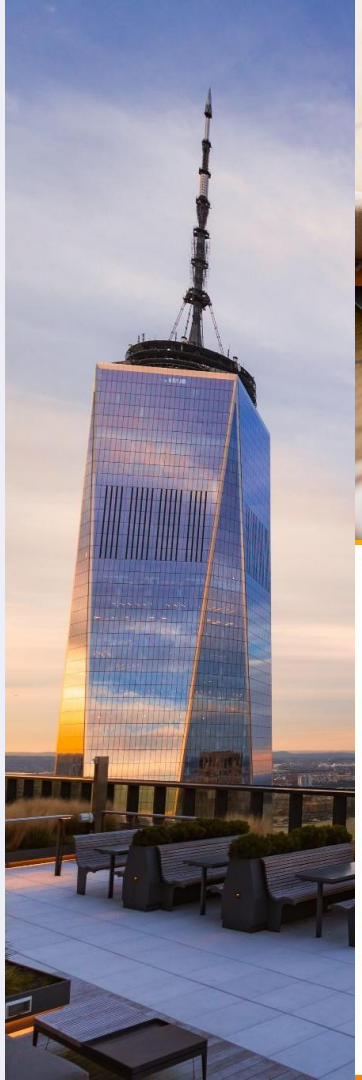
Platinum

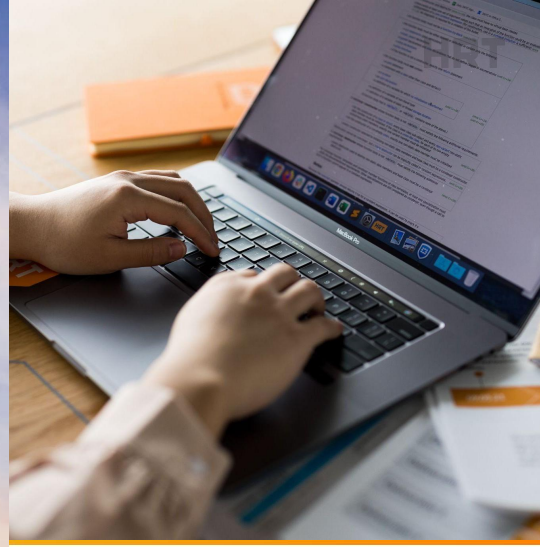


Hudson River Trading

Welcome!

hudson river trading





HRT

Meet the Team!



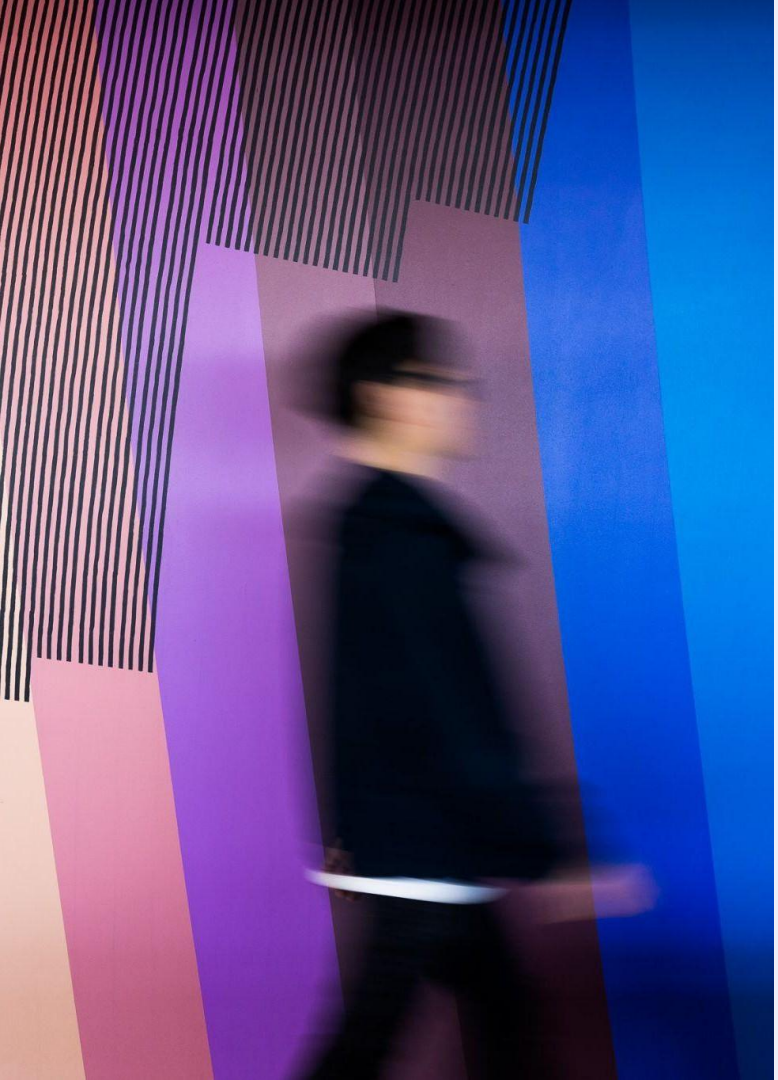
Joshan Parmar
Algo Developer



Ronette Yeboah
Recruiting Coordinator



Charlotte Whittington-Firth
EMEA Campus Lead



Agenda

01
Why HRT?

02
Join our team

03
HRT <> Oxhacks Challenge!



2. Why HRT?

Engineering Culture

almost all HRTers know how to code; our culture is technical, not financial

Highly Collaborative

open code base + open research environment

Technical Excellence

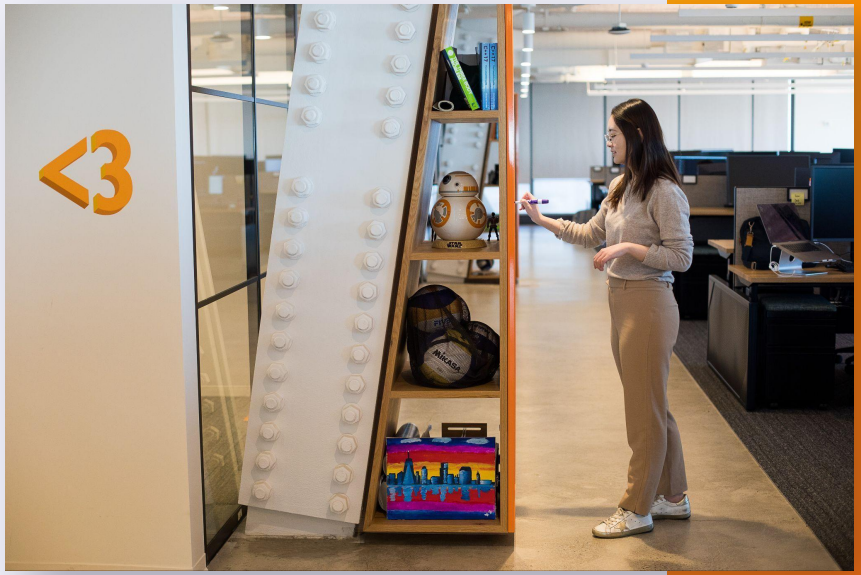
work with the best engineers and scientists

AI At Massive Scale

Data and compute at will

Engineering Culture

We're a **tech company** in finance, not a finance company in tech.

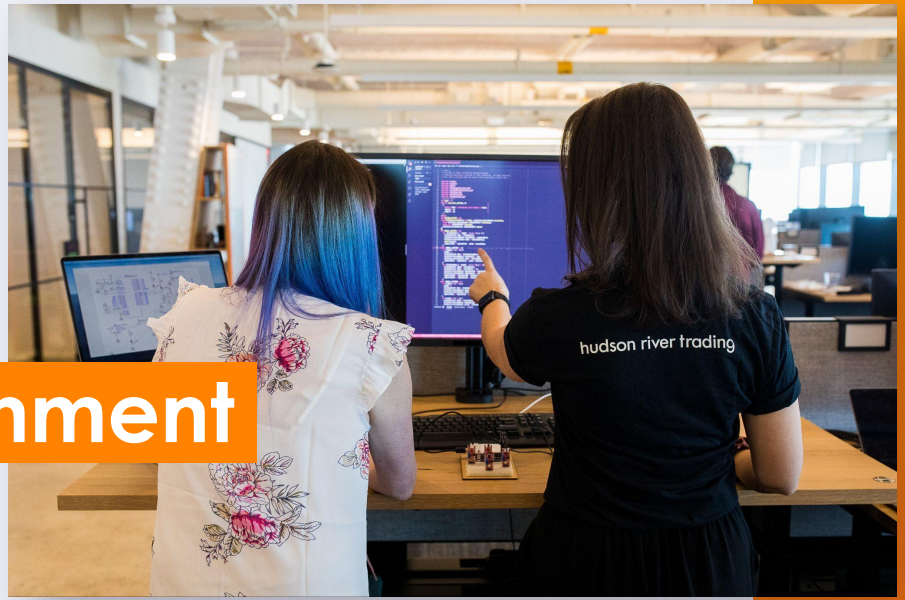


Highly Collaborative

open code base

+ open research

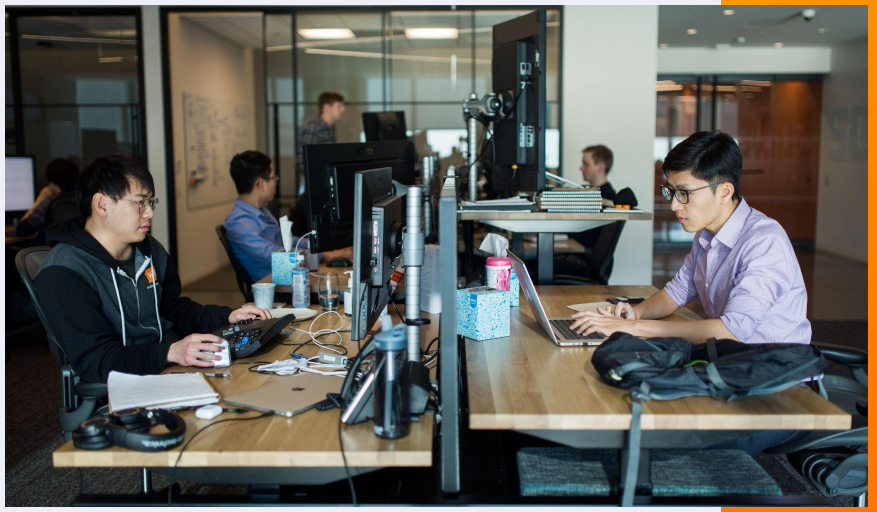
= **collaborative environment**



Technical Excellence

Make it Better

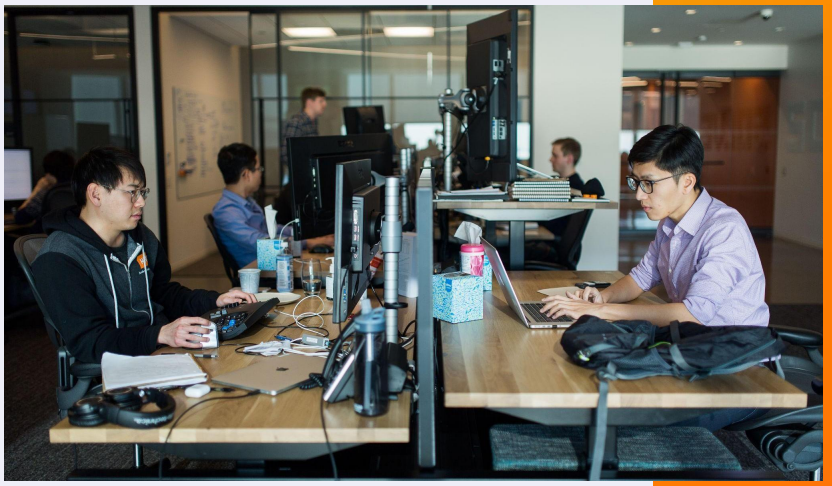
Small teams working on homegrown, on-prem infrastructure.



AI At Massive Scale

Experiment Without Limits

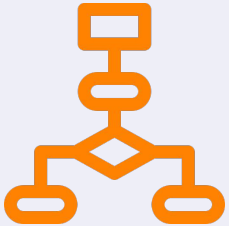
Industry-leading GPU and data infrastructure to unleash opportunities





3. Join our team

Roles at HRT



Algorithm Developers
Build and maintain the models that drive our trading.



Software Engineers
Create and maintain the infrastructure that is integral to our trading.



Data Scientist
Extract, validate, and normalize vast datasets to provide high-quality data for research and trading.

Algorithm Developers @ HRT



Algorithm Developers research, develop, and test novel order execution and model training methods to increase trading efficiency and maintain ongoing profitability.



A typical day involves applying rigorous statistical analysis to vast quantities of market and financial data to produce predictive trading models.



Areas of expertise:

Programming in Python and/or C++

Statistical analysis, numerical programming, or machine learning in Python, Pandas/Numpy.

Software Engineers @ HRT



Software Engineers are C++ and Python programmers whose work spans our incredible live trading system and a powerful research environment.



A typical day involves building tooling and infrastructure for our high-performance systems that is optimized for latency (live trading) and throughput (research).



Areas of expertise:

Python

C++

OS Internals

CPU Architecture

Networking Hardware & Protocols

Data Scientists @ HRT



Data Scientists focus on extracting useful information from the vast volume of data available in the world. They apply rigorous quality checks, and investigate methodologies for providing the highest quality data for research and trading.



A typical day involves analyzing multiple complex datasets, and developing scalable data pipelines to drive quantitative research across the firm.



Areas of expertise:

Python

Statistical Analysis & Machine Learning

Data Pipelines & Infrastructure Predictive

Modeling & Research

Culture of Problem Solving



Systems Architecture & Design

Cluster Management Large
Computing Complex
Systems Design



Performance Optimization

Software & Hardware Latency
Optimization Compression
Efficient Caching



Data & Computational Theory

Algorithms Game
Theory
Machine Learning



Information Technology

Databases
Wireless Communication
Compression
Big Data Analysis

2027 Student Opportunities @ HRT

Internships – Summer 2027

FOR 2028 GRADS

Algorithm Development (Quant Research) Intern

- Degree: Maths, CS, Physics, Stats, or related (open to undergrads & PhD students!)
- We are open to longer internships

Software Engineering Intern

- Degree: CS, Engineering, or related
- We are also open to extended placement students, as well!

Data Scientist Intern

- Degree: STEM field: *a computer science background is not required*

Full-time Roles

FOR 2026/2027 GRADS

Algorithm Developer

- Degree: Maths, CS, Physics, Stats, or related

Software Engineer

- Degree: CS, Engineering, or related

For all roles: You do not need finance knowledge or experience.

Interview Process

Step 01

Take Home Test

Full-time: ~2 hours timed

Internship: ~2 hours timed

Language: C++, Python

Step 02

First Phone

30 - 45 minute technical interview

SWE: Data structures, systems, algorithms, C and/or C++, and how computers work

Algo: Probability & Coding

Step 03

Second Phone

SWE: 1 hour technical interview via CoderPad. C/C++, primarily coding interview

Algo: 1 hour technical interview including, but not limited to Data analysis, programming, problem solving,

Step 04

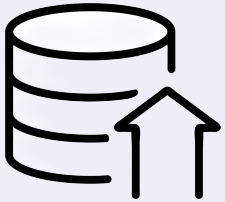
On-site

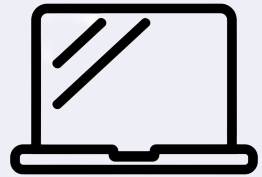
SWE: Assesses your programming skills, alongside your systems design knowledge

Algo: Including, but not limited Data analysis, programming & project work (C++ or Python)

HRT <> OxHacks Challenge 2026

Best use of Data for Predictions & Decision Making





Gold

OxHack 26



OxHack 26

Gold



SILVER

OxHack 26



SILVER

OxHack 26



OxHack 26

SILVER



SILVER

OxHack 26



Built with Delta

Show off how you worked together in Delta. Extra points for projects that solve real problems.

Work sync (or async) alongside teammates and agents. Use Delta from brainstorming/ideation, building, all the way through testing. Share your Delta threads to show examples of how you collaborated.

SILVER

OxHack 26



What We're Looking For

- Projects that feel **fast** - snappy UX, zero unnecessary friction.
- Projects that **solve a real problem**. Not a demo. Something you'd actually open tomorrow.
- Projects that are **thoughtful**. Clean code, smart architecture, good decisions.

Prompts to Get You Started

- What slows you down when you're coding? What if that friction disappeared?
- What's broken about collaborating on code with teammates or AI?
- What would you build if latency wasn't a constraint?
- What context do you lose every time you switch tasks — and how do you get it back?

SILVER

OxHack 26



How You Win

Speed & Performance (30%) — Does it feel fast? Is the UX snappy and responsive?

Technical Implementation (25%) — Clean code, smart architecture, good use of collaboration features.

Innovation (25%) — Fresh idea. Real problem. New angle.

Polish & Usability (20%) — Would you actually use this? Does it delight?

SILVER

OxHack 26



- **Download Delta** `https://delta.dev/download`
- **Tokens:** Use OpenRouter tokens in Delta. Docs here: https://delta.dev/docs/agents/models-and-providers#add-an-api-key
- **Quickstart:** https://delta.dev/docs/getting-started/quick-start
- **Useful videos**
  - Overview: https://www.youtube.com/watch?v=HWB48DFk4Sg
  - Subthreads: https://www.youtube.com/watch?v=yek9A_f39qo
  - Subthreads again: https://www.youtube.com/watch?v=i1IProLb_UM
  - Setting up Ollama: https://www.youtube.com/watch?v=Y4mpyFDb4xc
  - Not a demo but for CRDT nerds: https://www.youtube.com/watch?v=yif2rday2bw

Bronze

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TALOS

Partners

Partners

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MLH
MAJOR LEAGUE HACKING

HackathonsUK

Education

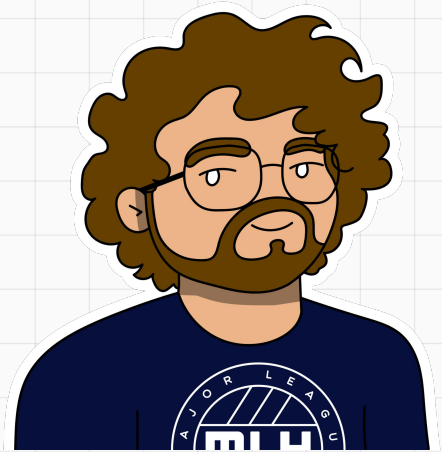
Partners

OxHack 26

MLH
MAJOR LEAGUE HACKING



Home of the next billion software creators.



@MajorLeagueHacking

@MajorLeagueHacking

@MLHacks

Hey! I'm Oscar.

I'm your Coach from Major League Hacking (MLH) this weekend. Fun facts about me:

- Studying **MEng Computer Science**
- I've been to **35 hackathons across 7 countries!**
- My favorite project I've made **was a painting with screens for eyes that follow people around a room!**

Everything you need this weekend:

mlh.link/oxhack

MLH

You can't learn the skills to become a **Software Creator** in the classroom.



Yashika Sharma
@yashika51

The MLH Community is the best thing that happened to me and my career ❤️



Brandon Barker
@projectbarks

MLH helped me become a better engineer than any internship or school ever could.



Emily Huang
@huang-emily

MLH taught me skills that set me apart from my peers & led to an amazing job at Microsoft!



Nigel Brown
@pnbrown

The skills & network I gained through MLH helped me land a great job at IBM.



Jessie Nguyen
@jessie_anh_ng

This summer was one of the most life changing experiences for me thanks to MLH.

MLH

The world's largest developer community.

Nearly **15% of all Software Engineers** got their start here in the MLH Community.

Through hackathons, internships, content, & more, MLH helps you **learn the skills & technologies you need** for your career.



OCTOBER 2026 • IN PERSON + ONLINE

Hacktoberfest 2026

AI belongs to everyone

Explore Hacktoberfest →

hacktoberfest.com



Powered by MLH + DEV • Presented by DigitalOcean

What is MLH doing here this weekend?

Resources for your hack:

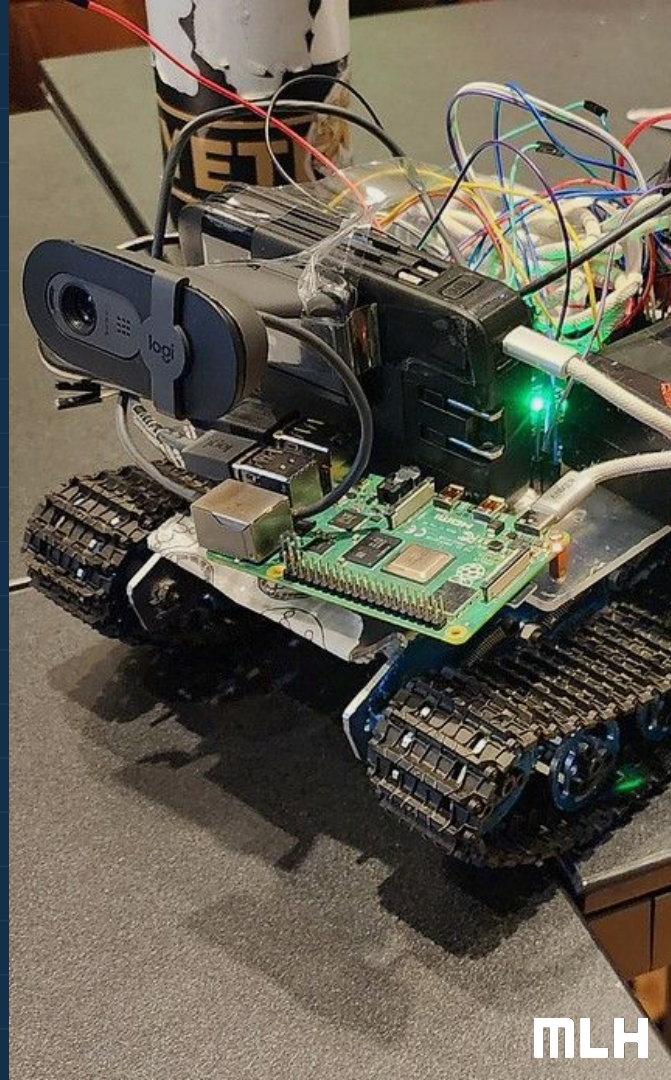
hack.mlh.com

MLH

MLH Hardware Lab

Check out some hardware for your project. Arduino, Raspberry Pi, and much more!

👉 Get Started: mlh.link/hardware



Intro to Google AI Studio

1. Grab an API Key in a flash at mlh.link/aistudio
2. Join the workshop at 17:30 in Classroom 1

Google Cloud



mlh.link/oxhack-aistudio

MLH

Hacking with GitHub Copilot

1. Fast-track access to the GitHub Student Developer Pack at mlh.link/pack
2. Exclusive MLH + GitHub Copilot sticker at 19:00 in Classroom 1



mlh.link/**oxhack-github**

MLH

2027 Season Mascot Tees!

Grab a limited edition t-shirt while supplies last 🐨👕



We have all the resources you need & some amazing prizes!

Resources for your hack:

hack.mlh.com

MLH

BEST USE OF GEMINI API

Unlock superpowers with Gemini's multimodality and flash models

🏆 You'll Win: MLH + DEV Swag Bags

👉 Get Started: mlh.link/gemini



BEST USE OF ELEVENLABS

==========

The world's best AI platform for producing & consuming audio.

🏆 You'll Win: Beats Solo Earbuds

👉 Get Started: mlh.link/elevenlabs



BEST USE OF TIGER DATA

Supercharge your hackathon project with the most scalable Postgres.

🏆 You'll Win: Stream Deck Mini

👉 Get Started: mlh.link/tigerdata



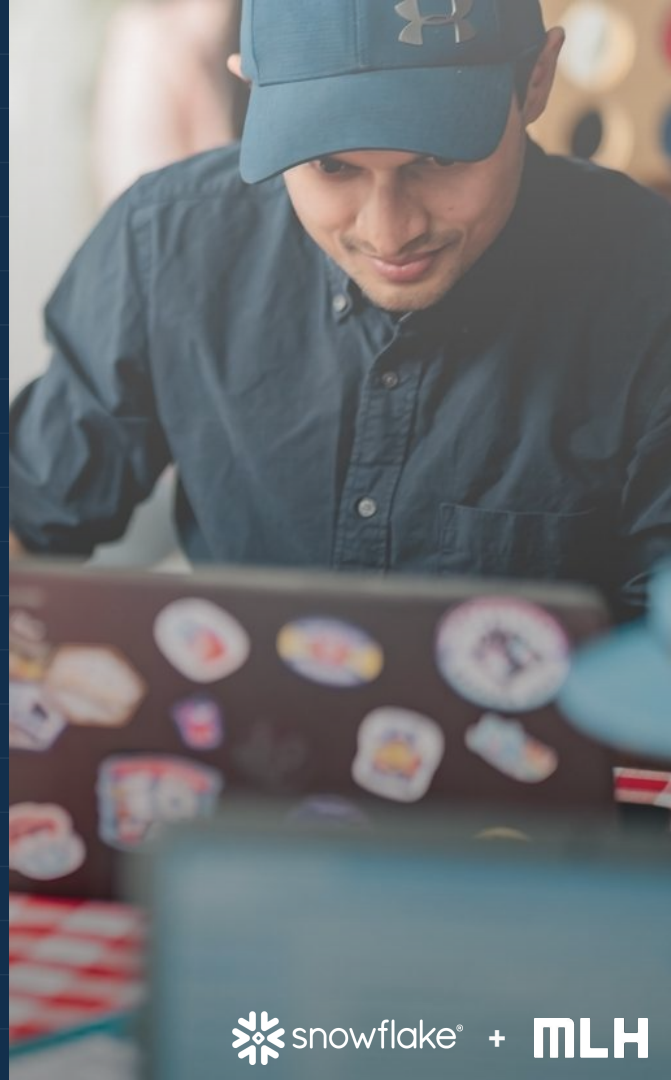
BEST USE OF SNOWFLAKE API

==========

Turn big data into big ideas with one powerful API.

🏆 You'll Win: Arduino Kits

👉 Get Started: mlh.link/snowflake



BEST USE OF SOLANA

Build high-speed apps on the world's fastest blockchain.

🏆 You'll Win: Assorted Prizes

👉 Get Started: mlh.link/solana



BEST USE OF PRESAGE

==========

Turn any camera into a real-time biometrics sensing device.

🏆 You'll Win: Fitbit Inspires

👉 Get Started: mlh.link/presage



BEST USE OF VULTR

==========

The ultimate developer toolkit for building on the cloud.

🏆 You'll Win: Portable Screens

👉 Get Started: mlh.link/vultr



DEV Writing Challenge

Your hackathon may end after the weekend, but your project story is just getting started. Publish what you built on DEV and help other developers learn from your process.

🏆 **You'll Win:** $100

👉 **Get Started:** dev.to/mlh-hackathon



DEV + MLH

Need to see those again?

mlh.link/oxhack

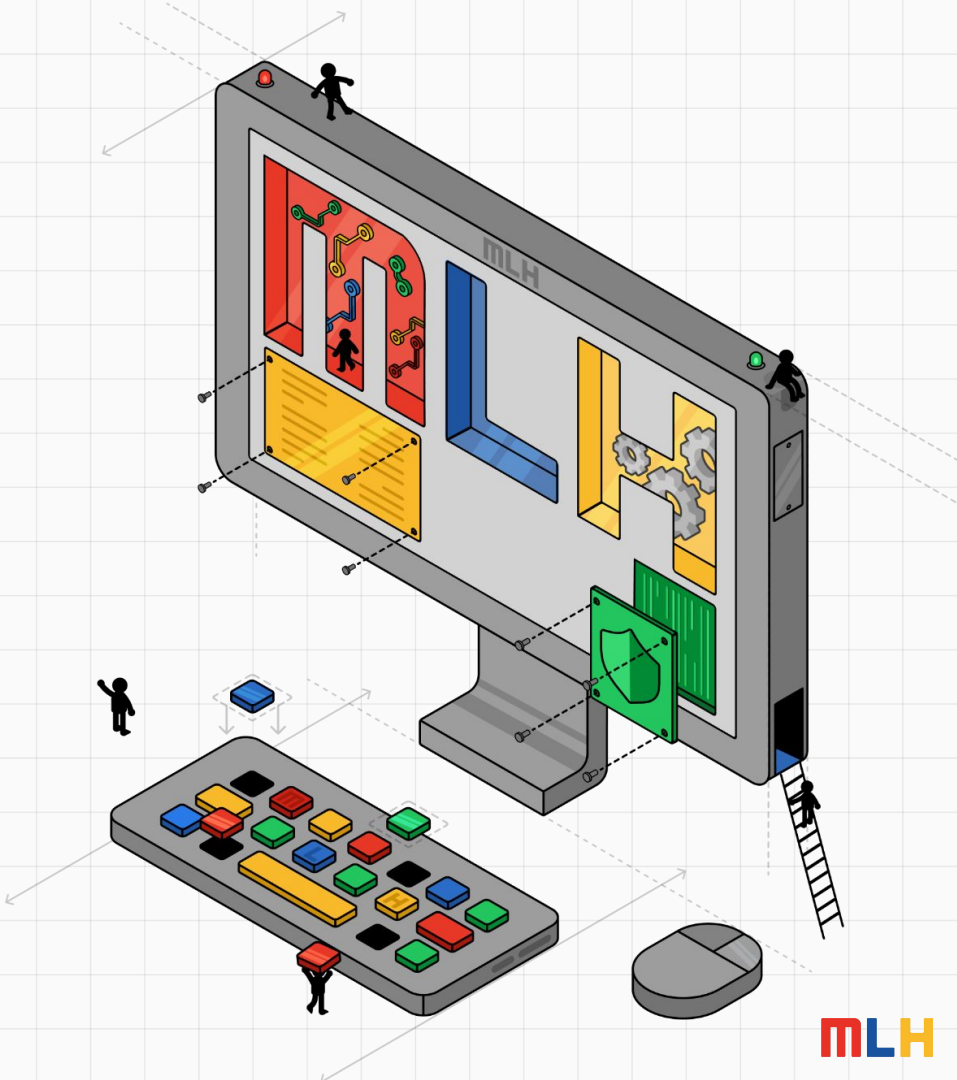
MLH

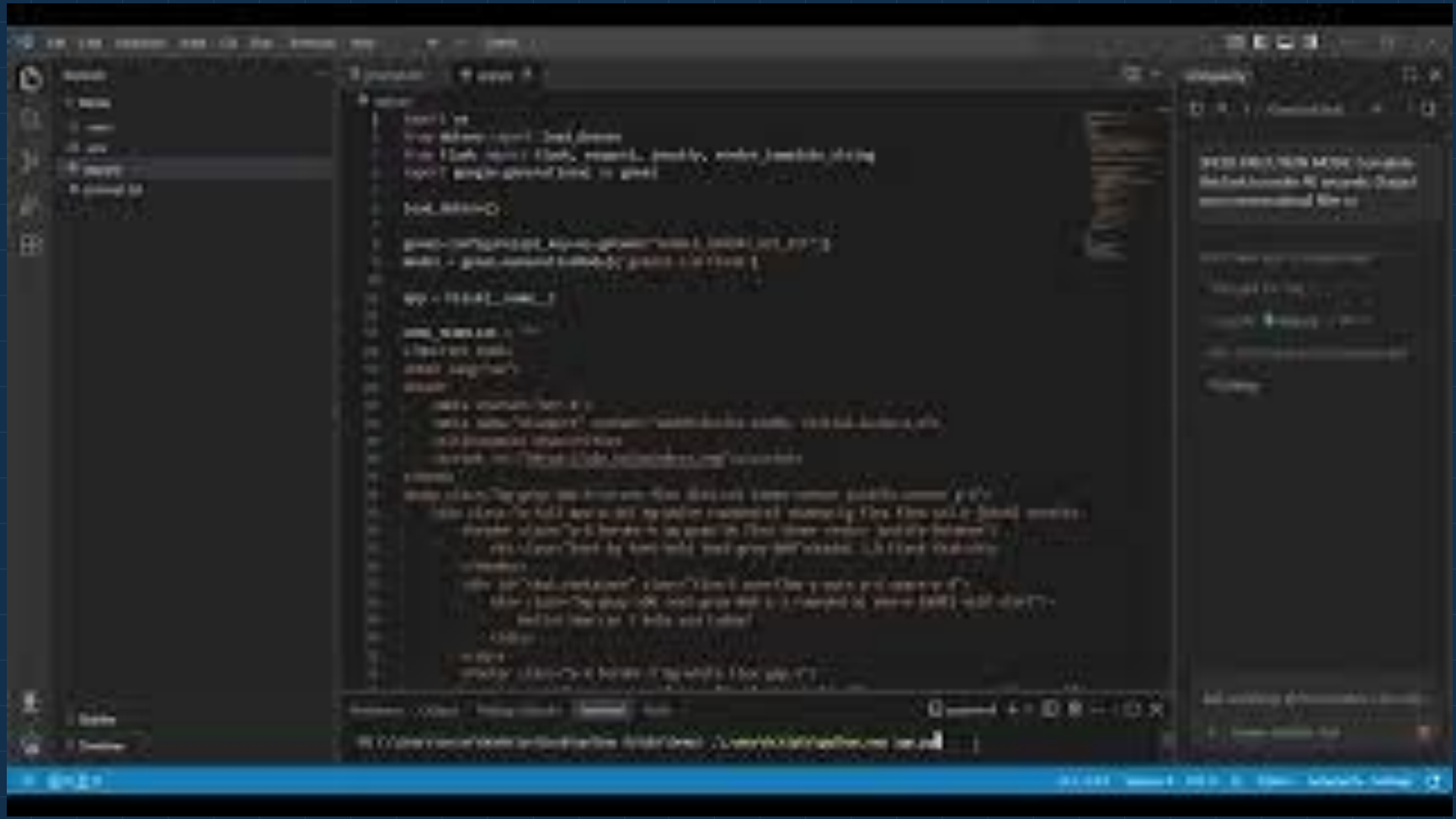
I Demoed Stickers

Earn your sticker by showing off what you build on Sunday

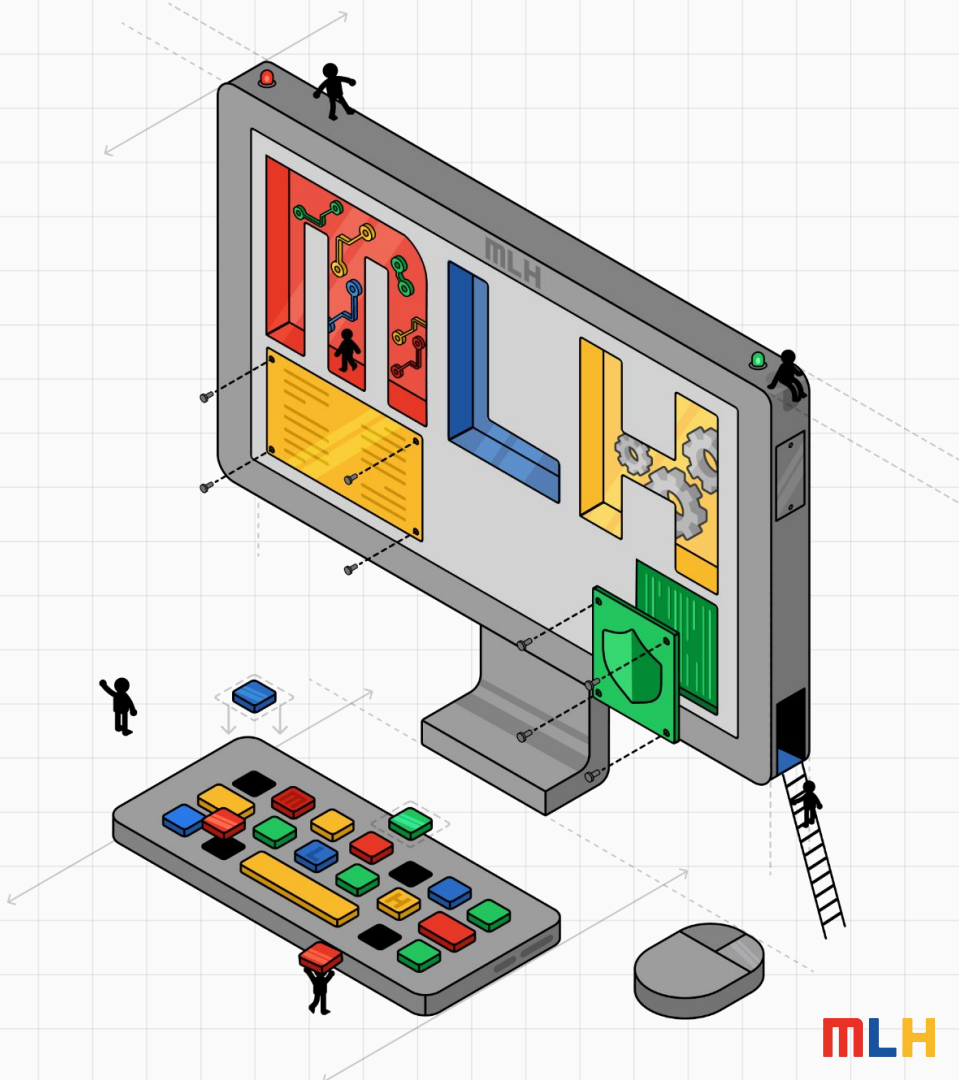


Let’s do a quick demo





mlh.link/
oscar-demo



On a serious note...

MLH Hackathons are safe & welcoming spaces for everyone:

`mlh.com/code-of-conduct`

Email: incidents@mlh.io

Phone: +44 (0) 800 808 5675

MLH



Home of the next billion software creators.

Partners

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HackathonsUK

Who are *Hackathons UK*?

Hackathon community in the UK





❯ Organise a hackathon

❯ Attend
a hackathon

hackuk.org/events





❯ **Mentor** at a hackathon

❯ Sponsor
a hackathon





› Join our **community**

What are we getting up to?

❯ Show off your **hacking skills**!

Hackiest Hack



❯ Show off your **stacking skills**

Cup Stacking

Saturday
20:30 BST



› Visit our table for lots of

Hackathons UK

› swag!




Hackathons UK
hackathons.org.uk

Partners

OxHack 26



GitHub Education

Brought to you by your GCE: @KaalkidanSahele




https://gh.io/oxhacks26
Get access to your student developer pack

Day One

9:00 - 11:00	Registration	
11:00 - 12:00	Opening Ceremony	L1
12:00 -	Hacking begins	Open hacking space
12:00 - 12:15	Team Building	
13:00 - 14:00	Lunch	Food Collection Point
14:00 - 15:00	Citadel Bubble Tea Drop	Food Collection Point
15:00 - 16: 00	HRT Talk from Matt Godbolt: What 25 Days of Optimizations Taught Me About Compilers	L1
17:30 - 18:00	MLH Workshop 1	C1
18:00 - 19:00	Dinner	Food Collection Point
19:00 - 19:30	MLH Workshop 2	C1
20:30 - 21:00	HackathonsUK Mini Event	C1
00:00 - 00:30	HRT Midnight Pizza	Food Collection Point

Day Two

8:00 - 9:00	Breakfast	Food Collection Point
9:00 - 10:00	Devpost Submission Workshop	C1
12:00	Hacking Finishes	
12:00 - 13:00	Lunch	Food Collection Point
13:00 - 15:45	Judging	All hacking spaces
16:00 - 16: 45	Closing ceremony	L1
16:45 - 17:00	Final group photo	

What about Daylight Savings?

- 25 hours of hacking!

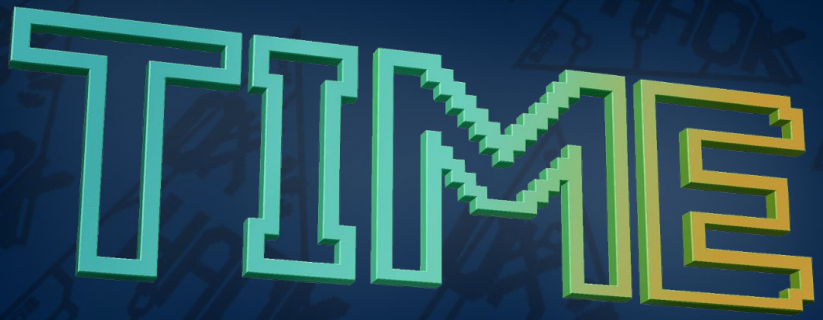
0xHack 2026



Build something, learn something, meet people and have fun!

0xHack 2026

Theme

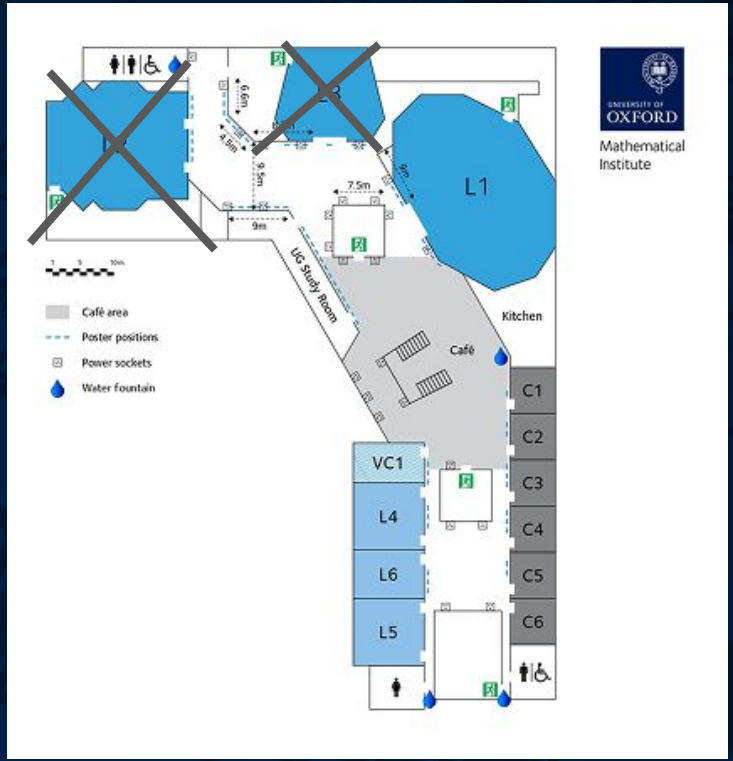


How to hack?

- Get creative, learn something new
- **“Best Handmade Project”** (With Thanks to Citadel)

Venue

L1	Ceremonies & Talks
C1	Mini events & workshops
C2 & C3	Hacking space
C4 & C5	Sleeping & resting space
C6	Organiser room
L4	HRT dedicated room
L5	Quiet hacking space
L6	Hacking space
Cafe area (and other open spaces)	Hacking spaces & Food collection point



Fire Safety Information

- No planned drills throughout the weekend
- In the case of a fire
  - Follow the organisers (OxHack Fleeces)
  - Assemble by Core 1 & Core 2
  - Follow Fire Exit arrows



Code of Conduct

- MLH Code of Conduct
  - https://github.com/MLH/mlh-policies/blob/main/code-of-conduct.md
- Hackathons UK
  - https://www.hackathons.org.uk/code-of-conduct/

Other notices

- Don't use the AV systems
- Accessibility needs
- First Aid marshalls at reception in case of emergency/incident
- Stick to venue areas
- Discord announcements channel

Happy Hacking

